SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA0325	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	K. Venkata Karthik	**Reg. No.:**	192372092
Section / Batch:	BE-CSE(AI)	**Date:**	12-10-2026

Code of Conduct

I K. Venkata Karthik / 192372092 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Venkata Karthik

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	4	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	6	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	4	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	4	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	2	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

S.No	Register No.	Student Name	Question
24	192372092	KOKKILIGADDA VENKATA KARTHIK	Design a concept map that evaluates duplexing and spectrum pairing in cellular systems. • Compare FDD and TDD, including uplink–downlink pairing, guard bands and traffic asymmetry. • Assess which duplexing method suits data-heavy mobile traffic better. • Support your conclusion through logical concept relationships.

Concept Map: Evaluation of Duplexing and Spectrum Pairing in Cellular Systems

Register No.: 192372092 Student Name: K. Venkata Karthik Question No.: 24

1. Aim and Core Idea

A cellular link must carry traffic in two directions: downlink (DL, base station to mobile) and uplink (UL, mobile to base station). Duplexing is the technique that keeps these two directions from interfering with each other. Spectrum pairing describes how radio spectrum must be allocated to make that possible: Frequency Division Duplex (FDD) needs a pair of separated bands, while Time Division Duplex (TDD) needs only one unpaired band that UL and DL share in turns.

The mobile network components involved (CO1) are the mobile device (UE), the base station with its antenna tower, and the radio channel between them. The duplexing choice decides how the base station and the device share that channel. The concept map below links the two methods through three comparison criteria (UL–DL pairing, guard mechanism and traffic asymmetry) and then evaluates which method suits data-heavy mobile traffic.

2. Duplexing in a Real Cell Site

Figure 1 shows a typical macro cell site: a lattice tower carrying three sector antennas serves mobile devices in the surrounding buildings and streets. On the left, FDD uses two separate carriers, so the DL and UL arrows exist together. On the right, TDD uses one carrier, and the same radio path carries DL slots and UL slots in alternation.

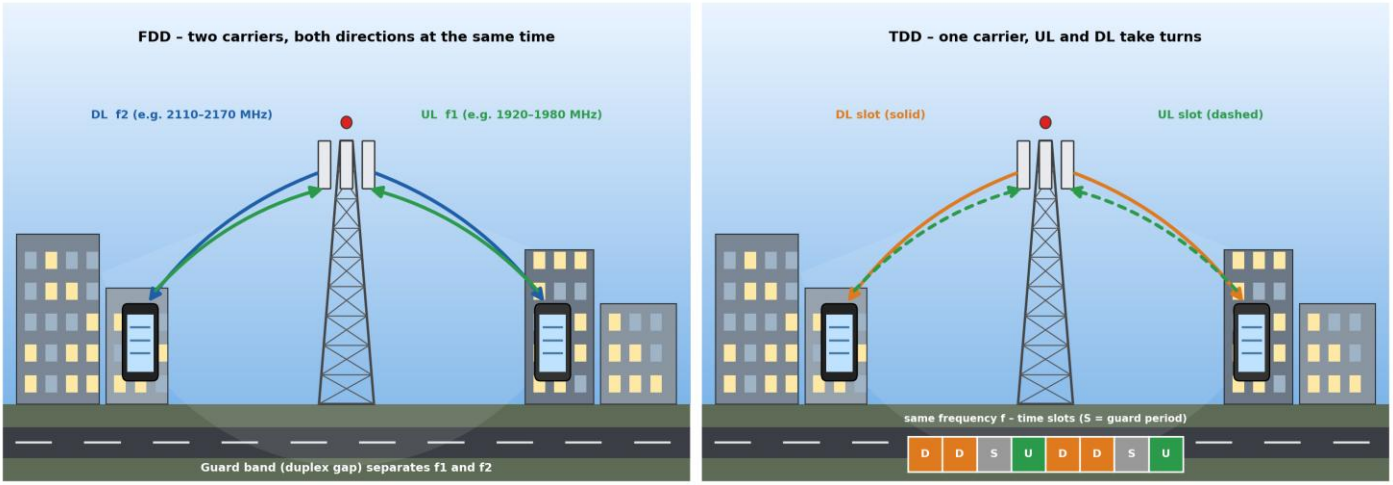


Figure 1: Cell-site illustration of FDD (two carriers, simultaneous) and TDD (one carrier, time slots)

3. Concept Map 1: FDD versus TDD

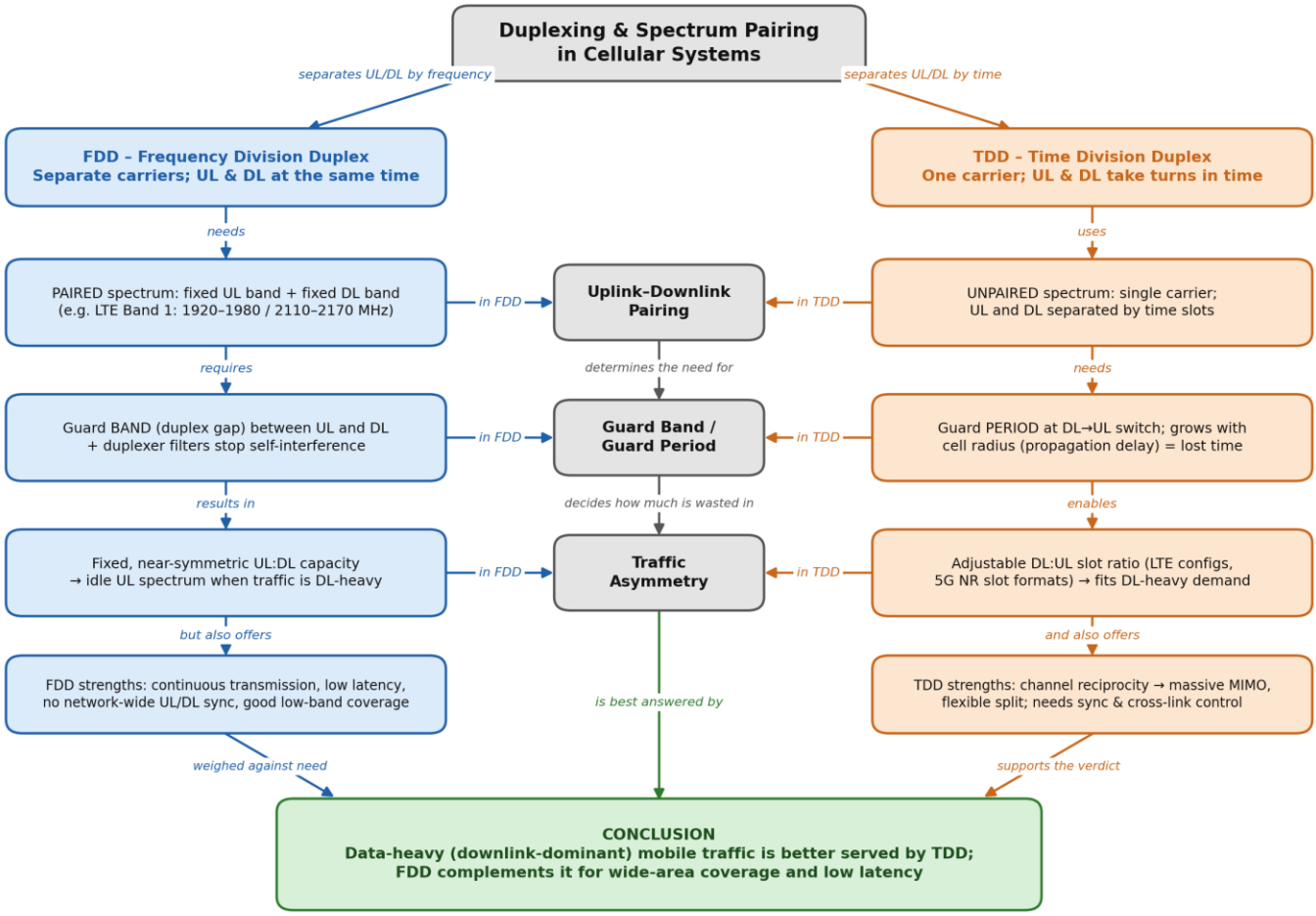


Figure 2: Concept map of duplexing and spectrum pairing (FDD in blue, TDD in orange, comparison criteria in grey, conclusion in green)

4. How the Spectrum Is Used: Frequency–Time View

Figure 3 shows the same two schemes on a frequency–time grid. FDD occupies two bands for all time, with an unused gap between them. TDD occupies one band, divided in time into DL and UL blocks, with a short guard period wherever the link switches from DL to UL.

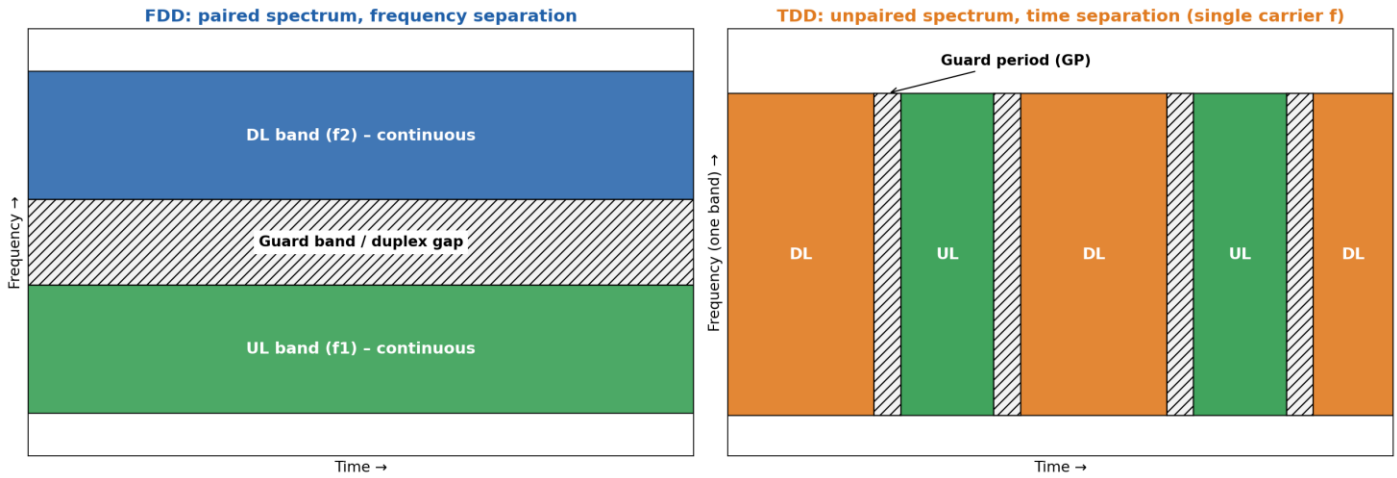


Figure 3: Frequency–time diagram of FDD (guard band) and TDD (guard period)

5. Comparison of FDD and TDD

Criterion	FDD	TDD
Basic principle	UL and DL transmit at the same time on two different carrier frequencies.	UL and DL transmit on the same carrier frequency in different time slots.
Uplink–downlink pairing	Needs PAIRED spectrum: a UL band and a DL band separated by a fixed duplex spacing (e.g. LTE Band 1: 1920–1980 MHz UL, 2110–2170 MHz DL).	Needs UNPAIRED spectrum: one contiguous band; no paired allocation or fixed duplex spacing is required.
Guard mechanism	Guard BAND (frequency): the duplex gap between the UL and DL bands, plus duplexer filters, stops the transmitter from leaking into the receiver.	Guard PERIOD (time): a short gap at the DL-to-UL switch that covers propagation delay and Tx/Rx switching time. It grows with cell radius, roughly 6.7 µs of round-trip delay per km, so it wastes more capacity in large cells.
Traffic asymmetry	UL and DL capacity are fixed by the widths of the two paired bands, so it is near-symmetric. When DL-heavy traffic arrives, the UL band sits partly idle while the DL band congests.	The DL:UL time ratio is configurable (LTE TDD has seven UL–DL configurations; 5G NR has flexible slot formats), so more slots can be given to the DL to match demand.
Spectrum availability	Paired blocks are scarcer and harder to allocate, and are mostly in lower bands.	Single blocks are easier to find, so TDD is widely used in mid-band and higher spectrum (e.g. 3.5 GHz).
Channel reciprocity / MIMO	UL and DL channels are on different frequencies, so the DL channel cannot be inferred directly from UL signals.	Same frequency in both directions, so the DL channel can be estimated from UL pilots. This supports beamforming and massive MIMO.
Latency and coverage	Continuous transmission in both directions gives low latency. Low-band FDD gives good wide-area and UL coverage.	A device may wait for its UL or DL slot, adding some latency. UL transmits only part of the time, so UL coverage is weaker than FDD at the same power.
Synchronisation and interference	Neighbouring cells need no UL/DL timing alignment.	Cells (and adjacent operators) must be time-synchronised and use compatible frame structures, otherwise cross-link UL-to-DL interference occurs.
Typical examples	LTE Band 1 (2100 MHz), Band 3 (1800 MHz: UL 1710–1785, DL 1805–1880 MHz), Band 8 (900 MHz).	LTE Band 40 (2300 MHz), Band 41 (2500 MHz), 5G NR n78 (3300–3800 MHz).

6. Traffic Asymmetry and the TDD Frame Structure

Traffic asymmetry means that DL and UL carry different amounts of data. Web browsing, video streaming and app downloads send small requests uplink and large content downlink, so DL demand is several times UL demand. LTE TDD handles this by choosing one of seven UL–DL configurations for each 10 ms radio frame. Figure 4 shows four of them; Configuration 5 gives eight DL sub-frames for every UL sub-frame, whereas Configuration 0 favours the uplink. The special sub-frame (S) holds the guard period. In FDD no such choice exists: the DL:UL capacity ratio is fixed by the paired bandwidths.

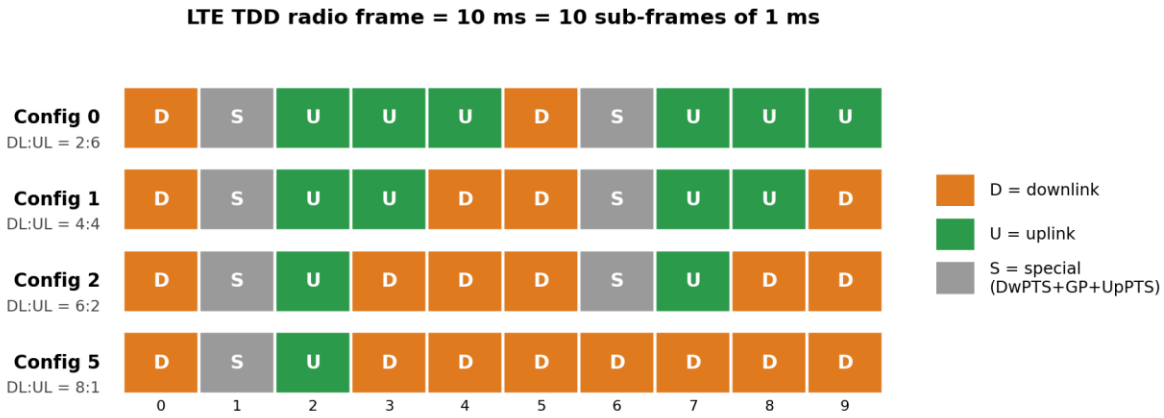


Figure 4: LTE TDD UL–DL configurations 0, 1, 2 and 5 (S = special sub-frame with guard period)

7. Worked Examples

Guard period: For a cell radius of 10 km, one-way propagation delay = 10 km / (3 × 10^8 m/s) = 33.3 µs, so the round-trip delay is about 66.7 µs. The guard period must be at least this long plus the Tx/Rx switching time. In LTE this is about one OFDM symbol (about 71 µs), and that time carries no user data. Doubling the radius doubles the guard period, which is why TDD guard overhead matters more in large rural cells than in small urban cells.

Spectrum utilisation: Assume 20 MHz in total and an offered load equal to the total capacity. FDD is split 10 MHz DL + 10 MHz UL (50:50). TDD uses the best-fitting LTE configuration. DL share of each configuration (special sub-frame ignored) is: Config 0 = 25 %, Config 1 = 50 %, Config 2 = 75 %, Config 5 = 89 %. Utilisation is the share of the spectrum that carries useful traffic, calculated as min(DL demand, DL capacity) + min(UL demand, UL capacity). The result is illustrative, since it ignores guard-period and signalling overhead.

Traffic (DL:UL demand)	FDD (50:50)	TDD (best config)	Observation
1:1 (balanced, e.g. voice)	100 %	100 % (Config 1)	No difference; FDD is just as good.
4:1 (browsing / social)	70 %	95 % (Config 2)	30 % of FDD capacity is wasted.
8:1 (video streaming)	61.1 %	100 % (Config 5)	FDD wastes about 39 % of the spectrum.

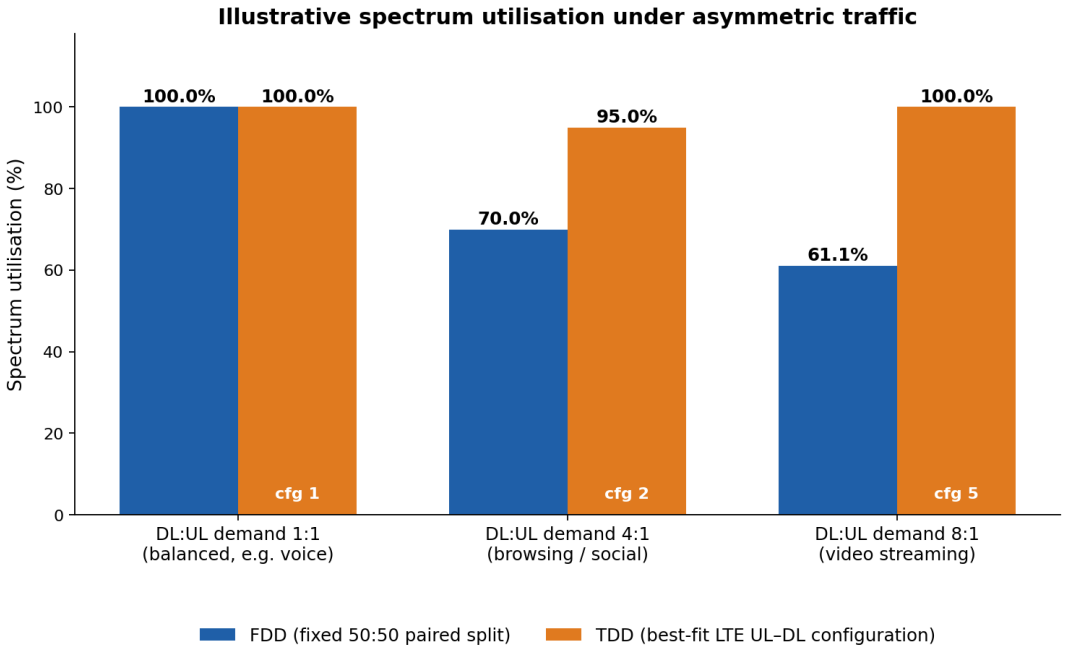


Figure 5: Illustrative spectrum utilisation of FDD and TDD as the traffic becomes more downlink-heavy

8. Concept Map 2: Evaluation for Data-Heavy Mobile Traffic

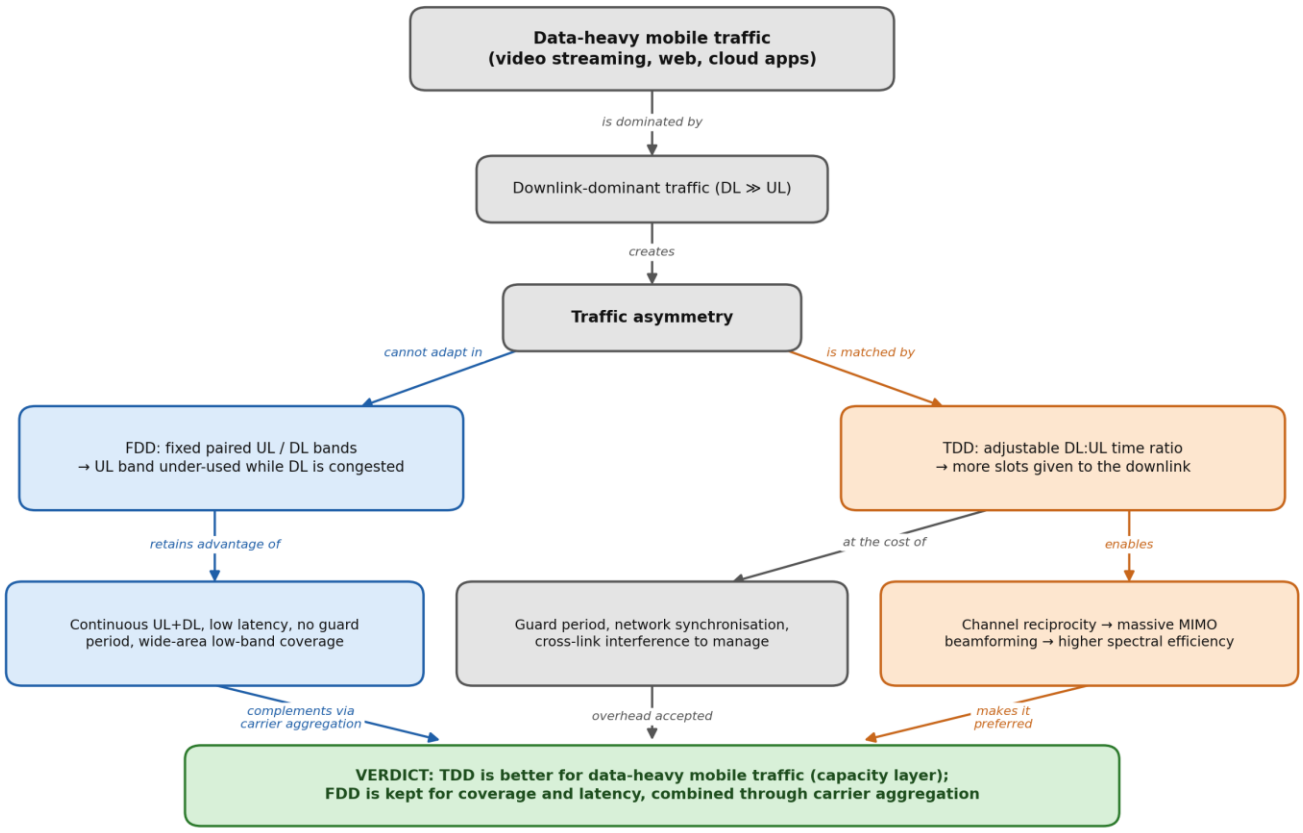


Figure 6: Logical chain from data-heavy traffic to the duplexing verdict

9. Linking Phrases and Logical Concept Relationships

Each arrow in the maps carries a linking phrase so that every pair of concepts reads as a proposition. The table below states and justifies the main ones.

From concept	Linking phrase	To concept	Justification
Duplexing	separates UL/DL by frequency in	FDD	Each direction gets its own carrier, so both can transmit simultaneously.
Duplexing	separates UL/DL by time in	TDD	One carrier is shared; UL and DL take turns in time slots.
FDD	needs	Paired spectrum	Two bands with fixed duplex spacing are required, one per direction.
TDD	uses	Unpaired spectrum	A single band is enough, which is easier to obtain.
Paired spectrum	requires	Guard band + duplexer	UL and DL are close in frequency, so filtering and a gap prevent self-interference.
TDD	needs	Guard period	UL and DL must not overlap in time despite propagation delay and switching.
Data-heavy traffic	is dominated by	Downlink traffic	Video, streaming, web and cloud content flow mainly from network to device.
Downlink-dominant traffic	creates	Traffic asymmetry	DL demand far exceeds UL demand.
Traffic asymmetry	cannot adapt in	FDD fixed split	Capacity per direction is fixed, so UL spectrum is under-used while DL congests.
Traffic asymmetry	is matched by	TDD adjustable ratio	Slots can be reallocated from UL to DL as demand shifts.
TDD	enables	Channel reciprocity → massive MIMO	Same-frequency operation lets DL beamforming use UL measurements, raising spectral efficiency.
TDD	at the cost of	Guard period, sync, cross-link interference	Overhead and coordination are the price of flexibility.
FDD	retains advantage of	Low latency and wide coverage	Continuous transmission and low-band propagation suit coverage and latency-critical service.

10. Qualitative Scoring for Data-Heavy Mobile Traffic

The table scores each method from 1 (poor) to 5 (excellent) on the criteria that matter for data-heavy traffic. The scores are a qualitative judgement from the relationships above, not measured values.

Criterion	FDD	TDD	Reason
Fit to downlink-heavy traffic	2	5	TDD can shift slots to the DL; FDD split is fixed.
Spectrum efficiency	3	5	No idle UL band; reciprocity supports massive MIMO.
Spectrum availability	2	4	Unpaired blocks are easier to obtain than paired blocks.
Latency	5	3	FDD transmits continuously; TDD waits for slots.

Criterion	FDD	TDD	Reason
Wide-area / uplink coverage	5	3	Low-band FDD and a full-time UL give better reach.
Synchronisation simplicity	5	2	TDD needs network-wide timing and interference control.
Total	20	22	TDD leads on the criteria that dominate data-heavy traffic.

11. Assessment and Conclusion

Assessment: Data-heavy mobile traffic (video streaming, browsing, app and cloud downloads) is strongly downlink-dominant. FDD gives each direction a fixed block of spectrum, so the uplink band is under-used while the downlink band is the bottleneck. TDD lets the operator shift the DL:UL ratio towards the downlink, so the same spectrum carries more useful data. TDD also gains from channel reciprocity, which supports massive MIMO and beamforming, and from unpaired spectrum that is easier to obtain in wider blocks.

Trade-offs: TDD pays for this with guard-period overhead (worse in large cells), strict network synchronisation, cross-link interference management and weaker uplink coverage. FDD remains stronger for wide-area coverage, low latency and balanced traffic such as voice. FDD operators also reduce its asymmetry problem with supplementary downlink carriers and carrier aggregation.

Conclusion: TDD suits data-heavy mobile traffic better, because its adjustable DL:UL split and reciprocity-based MIMO match asymmetric, downlink-dominant demand. In practice operators use FDD low bands for coverage and TDD mid bands for capacity, combined through carrier aggregation.